

Antareep Dey

Final Year Undergraduate
Computer Science & Engineering

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EDUCATION

- Vellore Institute of Technology, Bhopal Campus** Bhopal, India
Bachelor of Technology - Computer Science and Engineering 2022 - 2026
Relevant Coursework: Data Structures & Algorithms, Calculus & Laplace Transforms, Differential & Difference Equations, Applied Linear Algebra, Probability & Statistics, Discrete Mathematics & Graph Theory, Machine Learning

EXPERIENCE

- Indian Institute of Technology, Kharagpur** October 2025 - Present
Research Intern, Advisor: Dr. Abir Das
 - Worked on an **Image-to-Image synthesis** integrating **BLIP-2** captioning and **Stable Diffusion** for controlled synthetic data generation, **producing 650K augmented synthetic images** through **distributed inference**.
 - Proposed a **novel test-time adaptation** framework connecting video generation and TTA via **cross-modal synthesis**, leveraging generative priors to overcome target-domain data scarcity **without source data access**.
 - Designed a **teacher-student optimization** scheme with **action-conditioned video synthesis**, achieving **SOTA** across **3 benchmarks**.
- Vellore Institute of Technology, Bhopal Campus** August 2024- March 2025
Undergraduate Researcher, Advisor: Dr. Preetam Suman
 - Created a **medical dialogue corpus** through multi-source data aggregation and preprocessing of **379K patient-physician interactions**, enabling domain-agnostic medical natural language understanding capabilities across heterogeneous clinical contexts.
 - Fine-tuned a **3B parameter language** model using parameter-efficient methods such as **LoRA** achieving **59.05% accuracy** on medical benchmarks while **outperforming similar 7B parameter models** in resource-constrained environments. The Model and Dataset Opensource
 - Work accepted for **oral presentation at ICAIET 2025**
- Code In Place, Stanford University** April 2023 - June 2023
Student Instructor Remote
 - Taught a group of 15 students** from around the world through a 7-week course covering Python, console programming, and graphics programming
 - Prepared and **delivered 50-minute lessons** on a weekly basis based on **Stanford's CS106A** course

PUBLICATIONS

- Fine-Tuned Language Models in Healthcare: Empowering Affordable Medical Consultations** : Conference Paper with Oral Presentation at International Conference on Artificial Intelligence And Emerging Technologies, 2025

PROJECTS

- Explainable Musculoskeletal Fracture Analysis**(Work in progress): Researching a dual-pipeline framework for skeletal fracture detection, with a primary focus on **explainability in clinical settings**. Currently evaluating the efficacy of various **Class Activation Mapping (CAM)** techniques in producing interpretative heatmaps that align with radiographic gold standards. The ongoing work aims to bridge the gap between deep learning outputs and clinical decision-making through rigorous interpretability studies.
- CineGAN: High-Fidelity Cinematic Style Transfer using GAN's**  Developed a **Pix2Pix-based Generative Adversarial Network (GAN)** for cinematic style transfer, featuring a **ResNet generator** with **Squeeze-and-Excitation adversarial skip connections** and a **Multi-Scale Spectral PatchGAN discriminator**. Optimized **training on TPUs** using mixed precision and a composite loss function including **VGG19 Perceptual, Feature Matching, and Total Variation losses** to eliminate artifacts.
- SeeCurity: Surveillance Footage Optimization**  Engineered an intelligent surveillance system utilizing **MobileNetV2-SSD** architecture pre-trained on the COCO dataset for high-speed object detection. **Optimized for edge inference** to enable real-time inference on mobile devices, implementing a triggered-recording logic that captured video only upon detecting relevant entities to minimize storage overhead.

SKILLS

- Languages:** Python, C++, SQL, TypeScript, Bash
- Frameworks and Libraries:** PyTorch, Tensorflow, ScikitLearn, Pandas, Numpy, Seaborn, OpenCV, Flask, Astro, Bun, AWS, Weights and Biases, Transformers, Diffusers
- Soft Skills:** Leadership, Event Management, Writing

HONOURS AND AWARDS

- Raman Research Award-2026**, Vellore Institute of Technology, Bhopal Campus
- High Distinction Excellence-2020**, International Chemistry Quiz (Formerly ANCQ)